SANZHAR KIRGIZBAEV

SOFTWARE ENGINEER | APPLIED AI

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Software engineer building AI applications for real estate teams, from data processing and model integration to deployment. Work includes speech analytics, conversational assistants, and construction-document validation.

WORK EXPERIENCE

Tochno Construction Company | 2025 - Present
Software Engineer

- Built a sales-call analysis pipeline connecting Bitrix24 recordings, GigaAM transcription, and YandexGPT checklist scoring. Added personal-data masking and reports for reviewing individual calls and manager performance.
- Developed a real estate AI assistant with streaming responses, apartment search across synchronized XML feeds, mortgage and promotion information, and lead delivery to the CRM.
- Built a document-checking tool combining rule-based checks with model analysis of drawings and specifications. Matched AI findings against source text and linked issues to spreadsheet cells and drawing locations.
- Implemented voice training over WebSockets with streaming speech recognition and synthesis, configurable customer personas, and post-session scoring against company sales checklists.
- Built an internal marketing tool around 10,000+ apartment records, including pricing calculations, image generation, and an admin panel. Increased story production speed by approximately 3x.

Kvantland | 2025 - Present
Full-stack Developer / Team Lead

- Developed interactive problems with Nuxt.js, Flask, and PostgreSQL; contributed to five tournaments in mathematics, physics, and computer science.
- Promoted to Team Lead. Coordinated development, testing, and releases; assigned tasks, resolved blockers, and reviewed code.

Paywall RUS LLC | 2025 - 2026
Frontend Developer

- Developed responsive interfaces, fixed UI bugs, and completed assigned tasks ahead of release timelines. Participated in code reviews and release preparation.

Biamino | 2024 - 2025
Developer Intern

- Built investment-platform interfaces with Next.js and TypeScript, integrated Supabase, and managed server state with React Query.
- Developed a Telegram bot integrating Tesseract OCR and the Gemini API.

SELECTED AI & ML PROJECTS

Ship Detection & Ranging | 2026

Research prototype | Python, PyTorch, YOLO, OpenCV

- Fine-tuned a YOLO boat detector on reviewed test-basin footage. At confidence 0.45, precision improved from 0.740 to 0.832 and F1 from 0.616 to 0.670.
- Implemented boat tracking and distance estimation using depth maps or waterline geometry. Calibrated cameras against UWB and IMU telemetry and built tools for labeling and reviewing recordings.

HydroLink | 2026

Research prototype | Python, OpenCV, AprilTag, MQTT, Jetson

- Built a distributed perception pipeline across five shore cameras and two onboard cameras, connecting vessel and shore services through MQTT.
- Combined YOLO proposals with AprilTag decoding on Jetson Orin Nano devices. Added timestamp-aware capture, multi-device fusion, recording, and performance telemetry.

Sales Copilot | 2026

Prototype | Python, FastAPI, React, WebSockets, PostgreSQL

- Built a live sales assistant with three speech-recognition providers, hints linked to their sources, and feedback from sales managers.
- Added transcript replay for testing without speech-recognition costs. Instrumented p50/p95 time to hint and session cost; wrote offline scripts to compare ASR and LLM providers on the AMI meeting corpus.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI / ML: LLM APIs, speech recognition and synthesis, PyTorch, scikit-learn, OpenCV, YOLO, NumPy, SciPy

Backend & Data: FastAPI, Flask, Node.js, PostgreSQL, Redis, Qdrant, SQLAlchemy

Frontend: React, Next.js, Vue, Nuxt.js, Tailwind CSS

Infrastructure: Docker, Nginx, Git, WebSockets, SSE, MQTT

EDUCATION

Istanbul Technical University | 2024 - 2028 (expected)

Bachelor's degree in Artificial Intelligence and Data Engineering
Current GPA: 3.24 / 4.0

SELECTED ACHIEVEMENTS

2023 - Turkiye Burslari scholarship: tuition, accommodation, and monthly stipend.
2023 - 1st place, American University in Central Asia Mathematics Olympiad.
2022 - 3rd place, International Balatech Programming Competition.

Languages: English (Fluent, IELTS 7.5), Russian (Fluent),
Turkish (Advanced), Kyrgyz (Native)